Data Request: MaverickX Contamination Project

Triple Ignition Challenge team · Ishan Singh, Aisha Jawad, Beren Abedan, Trisha · 19 October 2026

Why we need it

We are asking for historical batch records, including confirmed contamination cases, so that we can test our contamination early-warning system on real MaverickX production. So far it has only been tested on simulated and public datasets, none of which contain contamination. With your data we can tell you how many hours earlier than your current checks it would have warned, how many false alarms you would have seen, and whether it could have traced shared sources such as a medium lot or a vessel.

What we are asking for

#	Data	Includes	Priority	Why
1	Process time series per batch	pH, temperature, dissolved oxygen, agitation, aeration, pressure, off-gas O_2/CO_2, batch start (inoculation) time	**Essential**	Learns normal behaviour and spots deviations
2	Controller outputs	Acid/base pump, feed pump, antifoam, cooling valve	**Essential**	Controllers can hide early signs; their effort still changes
3	Offline lab results	OD or biomass, enzyme activity, substrate, microscopy, plate counts, with sample time **and** result time	**Essential**	Uses each result only once it would really have been available
4	Batch outcomes (separate table)	Normal, contaminated or other failure; how confirmed; organism if known; when and how first noticed; final yield	**Essential**	Measures detection and lead time against your current method
5	Enough batches	At least 30 normal batches per product and process; every confirmed contamination case (ideally 10+); other problem batches	**Essential**	Defines normal variation reliably and tests for false alarms
6	Batch genealogy	Vessel, medium and raw-material lots, inoculum or seed lot, water, cleaning/sterilisation records, transfer lines, shift	**Important**	Traces shared sources across contaminated batches
7	Investigation records	Checks done, when, results, turnaround time, cost, timestamped operator notes	**Important**	Calibrates which check to recommend next
8	Sensor maintenance logs	Calibrations, probe replacements, known faults	**Useful**	Separates real deviations from faulty probes
9	Recipe and setpoints	Planned phase changes and additions	**Useful**	Stops planned events being flagged

Format, confidentiality and use

- **Format:** any CSV or Excel export from your existing systems. Please include units and the time zone. We handle the conversion, and bad readings are flagged, never deleted.
- **Confidentiality:** we will sign an NDA. Batch and lot IDs can be anonymised, provided the same lot keeps the same code. Values can be rescaled per variable, because the model works on deviations from normal, not absolute values.
- **Blind test:** if possible, keep back a set of later batches, including some contamination cases, and send their outcomes only after we report our results. This gives you an unbiased test.
- **Use:** earlier normal batches define "normal", and later batches are tested whole. The data is used only for this project and is deleted or returned at the end.

Questions for your scientists

☐ Which contaminants do you see most, and when do they usually start (after inoculation, a feed or sampling)?
☐ Which signals moved first in past contamination cases?
☐ At what hour is a batch no longer salvageable, and what can still be done before then?
☐ Could one scientist review our draft diagnosis rule book (about one hour)?

Contact: Ishan Singh, Triple Ignition Challenge team.